

RACHANA KUMAR

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SUMMARY

Computer Engineering graduate student with 2+ years of experience building full-stack, AI/ML, and embedded systems, including computer vision pipelines, distributed backend services, and cloud-deployed applications. Specializes in Python, Java, C++, and REST API development. Seeking a full-time Software Engineer or AI/ML Engineer position, available to start immediately.

EDUCATION

Arizona State University

August 2024 – May 2026

Master's of Computer Engineering (Computer Systems)

GPA - 3.93

Relevant Courses : Artificial Intelligence, Embedded Machine Learning, Cloud Computing, Data Mining, Data Processing at Scale.

Reva University

August 2019 – July 2023

Bachelor's of Technology of Electronics and Computer Engineering

GPA - 3.72

Relevant Courses : Data Structures and Algorithms, DBMS, Computer Architecture, Digital Electronics, Data science.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML, CSS, MySQL, PostgreSQL.

Cloud & Backend: AWS, REST APIs, Apache Kafka, Redis, Docker, Node-RED, WebSocket, MQTT.

Frameworks & Libraries: React, Angular, Spring Boot, TensorFlow, PyTorch, OpenCV.

Tools & Platforms: Git, GitHub, Linux, JIRA, Postman, Jenkins.

PROFESSIONAL WORK EXPERIENCE

Deepcept AI - AI Software Engineer

September 2023 – May 2024

- Developed Computer Vision pipelines for real-time intruder and object recognition using Python, OpenCV, and REST APIs.
- Integrated a remote access module into existing surveillance infrastructure via WebSocket and MQTT protocols, cutting operator response time, streamlining system accessibility, and boosting the overall operational efficiency.
- Built an end-to-end ML pipeline for Table Tennis stroke classification with TensorFlow/PyTorch, achieving real-time inference at 30fps and 92% recognition accuracy across all 8 defined stroke types on standard embedded hardware.

Telestream - Software Engineer

February 2023 – July 2023

- Contributed to backend development of PRISM, a real-time IP/SDI monitoring and analysis platform, processing high-throughput broadcast media streams for precise signal verification and quality compliance across all live channels.
- Delivered 20+ feature enhancements and resolved 90+ bugs across audio and waveform applications, cutting the open defect backlog by 40% and substantially improving overall product stability and long-term release confidence.
- Engineered C++ backend services and resolved frontend defects in JavaScript; collaborated in Agile/Scrum sprints using Git, JIRA, and Jenkins for sprint planning, bug tracking, and CI/CD release management across teams.

Bosch Ltd - Hybrid Tester Of Smart Solution Systems

July 2022 – January 2023

- Led QA across 5 of 14 smart campus subsystems, validating IoT integrations and automated control systems.
- Executed 300+ UAT test cases for mission-critical components, systematically documenting defects and coordinating with engineering teams to verify fixes and validate complete end-to-end functionality across all scheduled release cycles.
- Identified and escalated critical issues contributing to an 80% reduction in system downtime post-deployment.

PROJECTS

Mini-Uber Backend Architecture - Java, Spring Boot, Kafka, Redis, PostgreSQL

Spring 2026

- Designed a distributed system for ride-dispatch using Java/Spring Boot, Kafka message queues, and Redis geo-indexing; achieved sub-100ms driver matching latency under simulated load of 10k concurrent requests.

Multipurpose Camouflage Robot - ESP 8226 NodeMCU, Raspberry Pi, GSM Communication

Fall 2025

- Designed and operationalized a versatile camouflage robot employing an Embodied Semantic Perception model for adaptive concealment and strategic mobility.

Gym Exercise Tracker - Python, Mediapipe, Streamlit

Spring 2025

- Built a real-time exercise tracking app using Python, MediaPipe, and Streamlit, supporting 20+ exercise variations with automated rep-counting and form-alert notifications.

PUBLICATIONS

- "Camouflage surveillance system" at 2023 IEEE Fifth International Conference on Advances in Electronics, Computers, and Communications (ICAIECC), Bengaluru, India, DOI: 10.1109/ICAIECC59324.2023.10560310
- "Design and Development of Automated Loading Dock for **Local Anesthesia Deployment**"—Published in November 2022 at **Intellectual Property**, India - Application Number 202141047093.

ACHIEVEMENTS

- Published IEEE research on an AI-based surveillance system.
- REVA Hack** - Placed 9th among 650+ teams in the National Level Hackathon for "Gym Exercise Tracker".
- State Level - **KSCST** project funding for **Camouflage Surveillance System**.